

A Sharp Tail Barrier for Cucker–Smale Flocking

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Abstract

For the normalized Cucker–Smale flow we prove global existence, conserved mean velocity, exact variance dissipation, and a diameter comparison principle. The communication tail gives explicit confinement and exponential alignment, including unconditional flocking at the critical exponent $\beta = 1/2$. A scalar two-particle reduction proves the short-range threshold sharp and separates its zero-speed/unbounded-distance equality orbit from positive-speed escape. This round is the **global diameter-barrier owner**.

1 Model and all-particle barrier

For $N \geq 2$, $d \geq 1$, $K > 0$, and $\beta \geq 0$, set

$$\dot{x}_i = v_i, \quad \dot{v}_i = \frac{K}{N} \sum_{j=1}^N \psi(|x_j - x_i|)(v_j - v_i), \quad \psi(r) = (1 + r^2)^{-\beta}. \quad (1)$$

Write $\bar{v} = N^{-1} \sum_i v_i$ and $X = \max_{i,j} |x_i - x_j|$, $V = \max_{i,j} |v_i - v_j|$.

Theorem 1 (Global flow, dissipation, and tail barrier). *Equation (1) has a unique global classical solution. The mean velocity is constant and*

$$\frac{d}{dt} \frac{1}{N} \sum_i |v_i - \bar{v}|^2 = -\frac{K}{N^2} \sum_{i,j} \psi(|x_i - x_j|) |v_i - v_j|^2. \quad (2)$$

The upper Dini derivatives satisfy

$$D^+ X \leq V, \quad D^+ V \leq -K\psi(X)V. \quad (3)$$

If

$$V(0) < K \int_{X(0)}^{\infty} \psi(s) ds, \quad (4)$$

there is a unique $R \geq X(0)$ such that $K \int_{X(0)}^R \psi = V(0)$, and

$$X(t) \leq R, \quad V(t) \leq V(0)e^{-K\psi(R)t}. \quad (5)$$

Every $x_i - x_j$ converges and every v_i converges to $\bar{v}$. Consequently every initial state flocks when $0 \leq \beta \leq 1/2$.

Proof. For any direction, the largest velocity projection cannot increase and the smallest cannot decrease; hence the velocity convex hull is invariant. Velocities remain bounded, positions grow at most linearly, and local existence extends globally. Pair symmetry gives $\sum_i \dot{v}_i = 0$ and, after pairing (i, j) with (j, i) , gives (2) with its ordered-pair normalization.

If $V = 0$, all velocities coincide, every alignment acceleration vanishes, and the second inequality in (3) is immediate. Otherwise, at a velocity-diameter pair (p, q) put $e = (v_p - v_q)/V$ and $z_j = e \cdot v_j$. Then $z_q \leq z_j \leq z_p$. Every weight is at least $\psi(X)$, so

$$\dot{z}_p - \dot{z}_q \leq \frac{K\psi(X)}{N} \sum_j [(z_j - z_p) - (z_j - z_q)] = -K\psi(X)V.$$

Maximizing over active pairs proves the second Dini inequality; the identical active-pair argument for positions gives the first. Thus

$$D^+ \left(V + K \int_{X(0)}^X \psi(s) ds \right) \leq 0.$$

The radius R supplied by (4) is a first-crossing barrier. Since $\psi(X) \geq \psi(R)$, Gronwall yields (5). Integrability of V makes relative positions Cauchy, and conservation identifies the common velocity. Finally $\psi(s) \sim s^{-2\beta}$, so its tail diverges precisely for $\beta \leq 1/2$. $\square$

2 Explicit radius and sharp two-body boundary

This revision is the **critical-tail and scalar-sharpness owner**. Put

$$\Phi_\beta(r) = \int_0^r (1+s^2)^{-\beta} ds = r {}_2F_1(1/2, \beta; 3/2; -r^2).$$

The confinement radius is $R = \Phi_\beta^{-1}(\Phi_\beta(X(0)) + V(0)/K)$. In particular, $\Phi_{1/2} = \text{arsinh}$; the logarithmic divergence retains the endpoint $\beta = 1/2$ in the unconditional chamber.

Proposition 1 (Exact outward scalar trichotomy). *Take $N = 2$, $d = 1$, and orient the initial data so that $r(0) = r_0 \geq 0$ and $u(0) = u_0 \geq 0$. Then*

$$r' = u, \quad u' = -K\psi(r)u, \quad u(r) = u_0 - K \int_{r_0}^r \psi(s) ds. \quad (6)$$

For $\beta > 1/2$ let $A = K \int_{r_0}^\infty \psi$. If $u_0 < A$, then r converges to the unique finite zero of (6) and $u \rightarrow 0$. If $u_0 = A$, then $r \rightarrow \infty$ while $u \rightarrow 0$. If $u_0 > A$, then $u \rightarrow u_0 - A > 0$ and $r(t)/t \rightarrow u_0 - A$.

Proof. Subtracting the two equations gives (6); while $u > 0$, division by $r' = u$ gives $du/dr = -K\psi(r)$. Below threshold its strictly decreasing right side has one finite zero. At equality it stays positive at every finite r , so a finite limiting separation is impossible, while the tail tends to zero. Above threshold the positive limiting speed is $u_0 - A$, and averaging $r' = u$ proves the last limit. Thus the strict all-particle barrier is sharp on this subfamily, but its failure is not claimed necessary for general flocks. $\square$

3 Degenerate faces, evidence, and route boundary

This revision is the **boundary-atlas and scope-firewall owner**. For $N = 1$, both diameters vanish. At $K = 0$, velocities are constant and flocking occurs exactly when $V(0) = 0$. If $V(0) = 0$ for any $K \geq 0$, relative positions remain fixed. Coincident agents are regular because $\psi(0) = 1$. The two-body equality face is not classical flocking: velocity alignment holds but spatial confinement fails.

Exact rational rows audit mean conservation, the ordered-pair factor in (2), and active-diameter inequalities. Symbolic rows audit five communication primitives and the $\beta = 3/2$ below/equality/above threshold trichotomy. Finite evidence is regression only; the proof owns all N, d .

Cucker and Smale introduced this flocking model [1]; Ha and Liu developed the explicit Lyapunov-tail proof lineage [2]. We claim no priority. Nearby packages own fixed signed-Laplacian consensus, randomized gossip, and noisy Kuramoto synchronization, not this coevolving position-velocity flow.

The source parameters carry no rational-prime semantics, primitive-orbit ledger, dynamical zeta, target analytic bridge, or natural same-clock quantum lift. Route A is therefore rejected as $(A0_{\text{FAIL}}, A1_{\text{FAIL}}, A2_{\text{FAIL}}, A3_{\text{FAIL}}, A4_{\text{FAIL}})$. Route B remains locked under `NO_BAD_EULER_OR_ROOT_NUMBER`; no target zero match or Hilbert-Pólya operator is claimed.

References

- [1] F. Cucker and S. Smale, *Emergent behavior in flocks*, IEEE Trans. Automat. Control 52 (2007), 852–862. <https://doi.org/10.1109/TAC.2007.895842>.
- [2] S.-Y. Ha and J.-G. Liu, *A simple proof of the Cucker–Smale flocking dynamics and mean-field limit*, Commun. Math. Sci. 7 (2009), 297–325. <https://doi.org/10.4310/CMS.2009.v7.n2.a2>.